

# MAT 3150 — Homework 3

## Problem 4 Walkthrough: Averages of an Increasing Sequence

**Goal.** Let  $(x_n)_n$  be an increasing sequence of positive real numbers and define

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Prove that  $(y_n)$  is increasing as well.

### What “increasing” means

A sequence  $(a_n)$  is increasing if  $a_n \leq a_{n+1}$  for every  $n$ . (If your course uses *strictly* increasing, replace  $\leq$  by  $<$  everywhere below; the argument is the same.)

**The standard way to prove it:** show that

$$a_{n+1} - a_n \geq 0 \quad \text{for every } n.$$

With fractions, that means: put  $a_{n+1} - a_n$  over a common denominator, note the denominator is positive, and show the numerator is  $\geq 0$ .

**Notation that makes this problem manageable.** Let

$$S_n = x_1 + x_2 + \cdots + x_n, \quad \text{so} \quad y_n = \frac{S_n}{n}.$$

The one fact about  $S_n$  you’ll need:

$$S_{n+1} = S_n + x_{n+1}.$$

### Worked example: $a_n = \frac{n}{n+1}$ is increasing

**Compute the difference.** Common denominator  $(n+1)(n+2)$ :

$$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{n^2 + 2n + 1 - n^2 - 2n}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}.$$

**Sign check.** The numerator  $1 > 0$  and the denominator  $(n+1)(n+2) > 0$ , so  $a_{n+1} - a_n > 0$ .

*Proof.* For every  $n \in \mathbb{N}$ ,

$$a_{n+1} - a_n = \frac{1}{(n+1)(n+2)} > 0,$$

since the numerator and denominator are both positive. Hence  $a_n < a_{n+1}$  for every  $n$ , so  $(a_n)$  is increasing.  $\square$

**Pattern:** difference  $\rightarrow$  common denominator  $\rightarrow$  simplify  $\rightarrow$  show numerator and denominator have the right signs.

**Your turn:**  $y_n = S_n/n$

**1. Set up the difference.**

$$y_{n+1} - y_n = \frac{S_{n+1}}{\quad} - \frac{S_n}{\quad}.$$

**2. Common denominator.** The denominators are  $n+1$  and  $n$ , so use  $n(n+1)$ :

$$y_{n+1} - y_n = \frac{\quad \cdot S_{n+1} - \quad \cdot S_n}{n(n+1)}.$$

**3. Simplify the numerator.** Substitute  $S_{n+1} = S_n + x_{n+1}$  and expand. The  $nS_n$  terms should cancel:

$$\text{numerator} = \quad - \quad.$$

(*Target shape:* something times  $x_{n+1}$ , minus  $S_n$ .)

**4. Show the numerator is  $\geq 0$ .** This is where “increasing” is used.

- Since  $(x_n)$  is increasing, how does each of  $x_1, x_2, \dots, x_n$  compare to  $x_{n+1}$ ?

$$x_k \quad x_{n+1} \quad \text{for each } k = 1, \dots, n.$$

- Add those  $n$  inequalities together. What do you get for  $S_n$ ?

$$S_n = x_1 + \dots + x_n \leq \quad.$$

- So the numerator from step 3 is  $\geq \quad$ .

**5. Sign of the denominator.** Is  $n(n+1)$  positive? Why?

**6. The proof.** Fill in the template:

Let  $S_n = x_1 + \dots + x_n$ , so  $y_n = S_n/n$  and  $S_{n+1} = S_n + x_{n+1}$ . For every  $n$ ,

$$y_{n+1} - y_n = \frac{\quad}{n(n+1)}.$$

Since  $(x_n)$  is increasing,  $x_k \leq x_{n+1}$  for  $k = 1, \dots, n$ ; adding these gives  $S_n \leq \quad$ .

Hence the numerator is  $\geq 0$ . The denominator  $n(n+1)$  is positive. Therefore  $y_{n+1} - y_n \geq 0$ , i.e.  $y_n \leq y_{n+1}$  for every  $n$ , so  $(y_n)$  is increasing.

**Check yourself.**

- Where did you use that  $(x_n)$  is increasing?
- Did you use that the  $x_n$  are *positive*? (It's fine if you didn't — just notice.)